IAT New Test 7

Question paper, answer key and worked solutions

60 questions 180 minutes 240 marks

Biology 1–15; Chemistry 16–30; Mathematics 31–45; Physics 46–60.

Correct +4; incorrect -1; unattempted 0. All 60 questions are available.

This is a provisional 2026-pattern authoring draft for 2027 preparation. Independent academic review is pending.

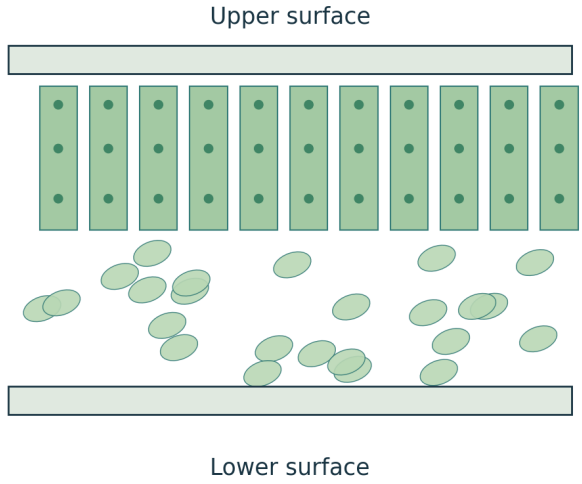
Instructions

Each item has exactly one correct option. Use the separate solution section for review after attempting the paper. **Figures and equations are part of the questions.**

Biology

Q1 An alga contains chlorophyll a and d plus phycobilins and stores floridean starch outside chloroplasts. Which group fits both clues?
A) Rhodophyceae
B) Phaeophyceae
C) Chlorophyceae
D) Bryophyta

Q2 In the illustrated leaf, elongated chloroplast-rich cells are below the upper epidermis; loosely packed cells with air spaces lie toward the lower side. Which anatomy fits?



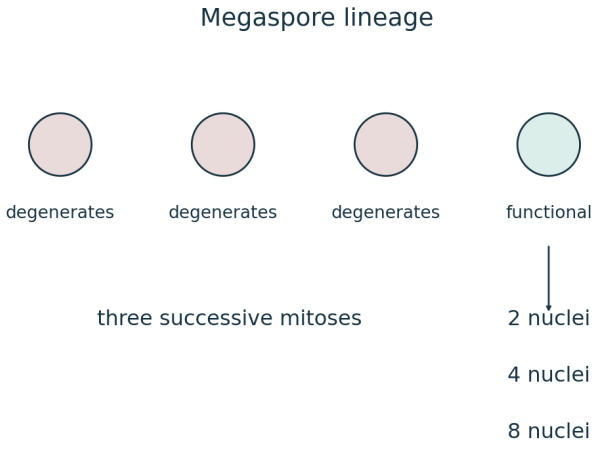
A) Isobilateral monocot leaf
B) Dorsiventral dicot leaf
C) Root cortex
D) Stem pith

Q3 A cell has duplicated DNA, but spindle attachment remains incomplete and sister chromatids do not separate. Which transition is blocked?
A) G_1 to S
B) S to G_2
C) Metaphase to anaphase
D) Anaphase to cytokinesis

Q4 At low light, extra CO_2 scarcely raises photosynthesis; after light is increased, the same CO_2 addition raises it. What changed?
A) CO_2 ceased to be a substrate
B) CO_2 destroyed chlorophyll
C) Respiration stopped
D) Light limitation was relieved, exposing CO_2 limitation

Q5 A previously sensitised person inhales pollen again and develops airway narrowing within minutes, with no infection. Which immediate mechanism fits?
A) Allergen cross-links mast-cell-bound IgE and releases histamine
B) Naive B cells finish a first response in seconds
C) Erythrocytes produce antibodies
D) Platelets secrete allergen-specific immunoglobulin

Q6 The diagram shows four meiotic megaspores, three degenerating, while the functional one undergoes three mitoses. How many nuclei are in the embryo sac before cellular organisation?



A) 4
B) 8
C) 16
D) 32

Q7 Two carrier parents have a child with an autosomal recessive disorder. Given that another sibling is unaffected, what is that sibling's carrier probability?
A) 1/4
B) 1/2
C) 2/3
D) 3/4

Q8 In a malaria model, treatment P removes circulating gametocytes but leaves asexual red-blood-cell stages initially intact. Treatment Q removes the asexual red-blood-cell stages but leaves gametocytes initially intact. Neither changes liver stages. Which immediate comparison correctly follows for fever-associated blood stages and infection of a feeding mosquito?
A) P immediately removes both fever-associated stages and mosquito infectivity
B) Q immediately removes both fever-associated stages and mosquito infectivity
C) P and Q act only inside the mosquito
D) P can reduce mosquito infection without immediately clearing asexual blood stages; Q can clear those stages while gametocytes remain

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Q9 A plasmid has an antibiotic marker and an insertion site within a colour reporter. A colony survives antibiotic yet loses reporter colour after ligation. What is supported?
A) Plasmid retention and possible insert disruption of reporter
B) No plasmid
C) Resistance gene destroyed
D) Insert orientation proven

Q10 Species A declines only when B is introduced; both use the same limiting seeds, and measured abiotic conditions remain unchanged. Which interaction is supported?
A) Mutualism
B) Interspecific competition
C) Commensalism
D) Predation necessarily

Q11 In a Hardy–Weinberg population, recessive aa frequency is 0.09. What heterozygote frequency follows?
A) 0.09
B) 0.21
C) 0.42
D) 0.49

Q12 A heritable insect colour variant exists before habitat colour changes. Birds then catch one phenotype more often, and that phenotype declines over generations. What supports selection?
A) Individuals recolour themselves
B) Both colours reproduce equally
C) Neither colour is inherited
D) Differential survival and transmission of the surviving colour

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Q13 At rest, heart rate is 75 min^{-1} and stroke volume 80 mL; arterial and mixed venous oxygen contents are 200 and 150 mL O_2 per litre of blood. During exercise, the corresponding values are 120 min^{-1}, 100 mL, 200 and 100 mL O_2 per litre. In steady state use oxygen consumption = cardiac output × arterial–venous oxygen-content difference. What are exercise/rest ratios of cardiac output and oxygen consumption?
A) 2 and 4
B) 2 and 2
C) 1.5 and 3
D) 2.5 and 5

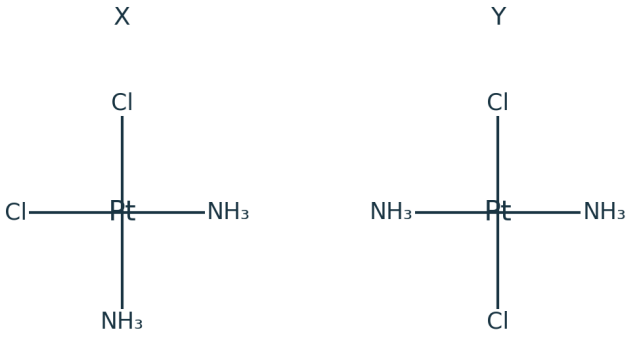
Q14 An enzyme hydrolyses peptide bonds in a protein but leaves the constituent amino acids intact. Which structural level is directly disrupted?
A) Only quaternary assembly
B) Primary amino-acid sequence
C) Only helix hydrogen bonds
D) Nucleic-acid phosphodiester chain

Q15 Six CO_2 fixed in a simplified Calvin-cycle account give twelve G3P. Two G3P leave as a six-carbon sugar equivalent. How many G3P return to regenerate six five-carbon RuBP?
A) 6
B) 8
C) 10
D) 12

Chemistry

Q16 Three equivalent carbonate-ion resonance drawings put C=O at different oxygen atoms. What stays fixed in every valid contributor?
A) Double-bond location
B) Nuclear positions and connectivity
C) Each oxygen's individual formal charge
D) Only total electron count changes

Q17 The figure shows square-planar $[Pt(NH_3)_2Cl_2]$ in two layouts: adjacent Cl ligands in X, opposite Cl ligands in Y. How are X and Y related?



A) Cis-trans geometrical isomers
B) Coordination-number isomers
C) Optical enantiomers
D) The same complex after rotation

Q18 Permanganate is reduced to Mn^{2+} in sufficiently acidic solution, but to MnO_2 in near-neutral conditions. What accounts for the contrast?
A) Mn has one oxidation state
B) Oxygen turns into nitrogen
C) Permanganate cannot oxidise in neutral solution
D) Medium-dependent pathways favour different Mn oxidation states

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Q19 Why do atomic radii generally decrease from Li toward F in the second period?
A) Nuclear charge increases while new electrons enter the same shell
B) An inner shell is added each step
C) Proton count is unchanged
D) Shielding fully cancels added protons

Q20 The figure lists four proposed electron quantum-number rows (n,l,m_l,m_s). Which row is forbidden?

Candidate quantum-number sets

	(n, l, m_l, m_s)
A	$(3, 2, -2, +1/2)$
B	$(2, 1, 0, -1/2)$
C	$(3, 3, 0, +1/2)$
D	$(4, 0, 0, -1/2)$

A) A
B) C
C) B
D) D

Q21 At equal temperature and pressure, 2.0 L H_2 reacts with 1.0 L O_2 to form water vapour without condensation. What vapour volume forms?
A) 1.0 L
B) 1.5 L
C) 2.0 L
D) 3.0 L

Q22 In ordinary addition of HBr to propene without peroxide, which species acts as electrophile in the first step?
A) Br^-
B) Propene π bond
C) A carbon radical
D) H^+ from polarised HBr

Q23 Propene reacts with HBr in trial X without peroxide and trial Y under peroxide radical conditions. What major products arise, X then Y?
A) 2-bromopropane; 1-bromopropane
B) 1-bromopropane; 2-bromopropane
C) Both 2-bromopropane
D) Both 1-bromopropane

Q24 Under polar protic conditions, compare substitution of tert-butyl bromide and methyl bromide by hydroxide. Which dominant limiting mechanisms are plausible?
A) Both SN1
B) Tertiary SN1; methyl SN2
C) Methyl SN1; tertiary SN2
D) Both only elimination

Q25 A primary and a secondary alcohol are separately oxidised only to their first carbonyl products. How do those products respond to Tollens reagent?
A) Both ketones, negative
B) Both aldehydes, positive
C) Primary aldehyde positive; secondary ketone usually negative
D) Primary ether; secondary ester

Q26 Sucrose is nonreducing before hydrolysis, but its hydrolysate reduces Fehling reagent. Why?
A) Hydrolysis destroys all anomeric centres
B) Fehling detects peptide bonds
C) Sucrose has no oxygen
D) Its bond ties both anomeric centres; hydrolysis frees reducing monosaccharides

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Q27 For monoprotic weak acid HA with $K_a = 10^{-5}$ and initial concentration 0.10 M, estimate pH using small dissociation.
A) 2
B) 2.5
C) 3
D) 5

Q28 At equal molality in the same solvent, a solute gives twice glucose's freezing-point depression. Assuming complete ideal dissociation, how many particles per formula unit form?
A) One
B) Three
C) Four
D) Two

Q29 A two-electron cell has $E^\circ = 1.10$ V and $Q = 100$ at 298 K. With $E = E^\circ - (0.0592/n)\log Q$, what is E?
A) 1.0408 V
B) 1.1592 V
C) 0.9800 V
D) 1.1000 V

Q30 If $\Delta H > 0$ and $\Delta S > 0$ are nearly constant, when can a reaction be spontaneous at constant pressure?
A) Only low T
B) High enough T that $T\Delta S > \Delta H$
C) Every T
D) Never

Mathematics

Q31 An ellipse has foci (±3,0) and major-axis vertices (±5,0). Find its eccentricity and right directrix.
A) 3/5 and x=25/3
B) 5/3 and x=3
C) 3/5 and x=5/3
D) 4/5 and x=25/4

Q32 The normal to $y^2=8x$ at (2,4) meets the parabola again. What is the second intersection point?
A) (18,12)
B) (6,0)
C) (2,−4)
D) (18,−12)

Q33 Evaluate $\int_0^1 dx/((x+1)(x+2))$.
A) ln(4/3)
B) ln(3/2)
C) ln2
D) 1/2

Q34 Evaluate $\int_{-1}^{1} |x|\,dx$.
A) 0
B) 1/2
C) 1
D) 2

Q35 The tangent to $x^2+xy+y^2=7$ at (1,2) crosses the x-axis at which coordinate?
A) 5/2
B) 3
C) 7/2
D) 4

Q36 For f:R→R defined by $f(x)=x^3-3x$, which is true?
A) Both injective and surjective
B) Injective only
C) Neither
D) Surjective only

Q37 An invertible 3×3 matrix A has det A=−2. What is $\det(2\,\mathrm{adj}(A^{-1}))$?
A) −2
B) 1/2
C) 2
D) 8

Q38 Events A,B are mutually exclusive with P(A)=1/2, P(B)=1/3. What is P(A|A∪B), and are they independent?
A) 1/2, yes
B) 3/5, yes
C) 2/3, no
D) 3/5, no

Q39 Use principal arctan values in $(-\pi/2,\pi/2)$. Evaluate $\arctan(\tan(2\pi/3))$.
A) $-\pi/3$
B) $2\pi/3$
C) $\pi/3$
D) $-2\pi/3$

Q40 Find shortest distance between lines $r=(t,0,0)$ and $r=(0,1,2)+s(0,1,1)$.
A) 1
B) $1/\sqrt{2}$
C) $\sqrt{2}$
D) 2

Q41 For $a=(3,4,0)$, $b=(2,-1,2)$, find vector projection of a onto b.
A) $(2/3)b$
B) $(2/9)a$
C) $(3/2)b$
D) $(2/9)b$

Q42 A monic cubic with real coefficients has roots 2 and $1+i$. What is its constant coefficient?
A) -2
B) -4
C) 4
D) 6

Q43 For which k does $x^2-2x-k=0$ have two distinct real roots strictly in $(0,3)$?
A) $-1<k<0$
B) $-1\le k\le 0$
C) $0<k<3$
D) $k>0$

Q44 Sum the two middle coefficients of $(1+x)^9$, then divide by the middle coefficient of $(1+x)^8$. What is the ratio?
A) 9/5
B) 18/5
C) 7/2
D) 4

Q45 An arithmetic-geometric progression has kth term $k2^{k-1}$. What is the sum of its first eight terms?
A) 769
B) 1537
C) 1793
D) 2049

Physics

Q46 A ball is thrown vertically upward at 20 m s^{-1}. For the first second its downward acceleration is 10 m s^{-2}; thereafter, in a simplified model, it is 5 m s^{-2} until the ball turns. How long after release does it first reach its highest point?
A) 2 s
B) 2.5 s
C) 3 s
D) 4 s

Q47 Two blocks of masses 2 kg and 3 kg lie on a smooth horizontal floor of an elevator and are joined by a taut string. A 10 N horizontal force pulls the 3 kg block; the elevator accelerates upward at 2 m s^{-2}. What is the string tension?
A) 4 N
B) 2 N
C) 6 N
D) 10 N

Q48 A 1 kg mass falls 0.20 m onto an uncompressed vertical spring of constant 200 N m^{-1}. Neglect losses and take g=10 m s^{-2}. What is maximum compression below first contact?
A) 0.10 m
B) 0.15 m
C) 0.30 m
D) 0.20 m

Q49 A solid cylinder rolls without slipping down a vertical drop of 1.5 m from rest. With g=10 m s^{-2} and $I = MR^2/2$, what is its centre-of-mass speed at the bottom?
A) $\sqrt{15}$ m s^{-1}
B) $2\sqrt{5}$ m s^{-1}
C) $\sqrt{30}$ m s^{-1}
D) $\sqrt{60}$ m s^{-1}

Q50 A satellite moves in a circular orbit of radius r around a fixed planet of mass M. It is moved to a circular orbit of radius 4r around the same planet. What is the ratio of new to old orbital period?
A) 2
B) 4
C) 8
D) 16

Q51 A uniform block of density 0.8 times water density and volume V is held so that half its volume is submerged. What is the tension in a supporting string from above, as a fraction of its true weight? Ignore surface tension.
A) 1/8
B) 3/8
C) 1/2
D) 5/8

Q52 A gas absorbs 300 J of heat and does 100 J of work during expansion. It is then compressed adiabatically with 50 J of work done on the gas. What is the total change in internal energy over both stages?
A) 150 J
B) 200 J
C) 250 J
D) 350 J

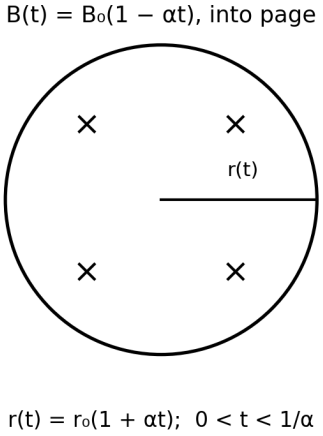
Q53 An electric dipole in a uniform field is rotated quasistatically from 60° to 120° relative to the field. How do its torque magnitude at the endpoints and its potential energy change compare?
A) Torque increases; energy unchanged
B) Torque decreases; energy falls by pE
C) Torque is unchanged; energy falls by pE
D) Torque is unchanged; energy rises by pE

Q54 The plate separation of a parallel-plate capacitor is doubled without changing plate area. Compare energy stored when (i) a battery keeps voltage fixed and (ii) the charged capacitor is disconnected before the change.
A) (i) halves; (ii) doubles
B) Both energies halve
C) (i) doubles; (ii) halves
D) Both energies double

Q55 Two long parallel wires carry currents I and 2I in the same direction, separated by d. Both currents are doubled and their separation is doubled. By what factor does force per unit length change?
A) 1/2
B) 1
C) 2
D) 4

Q56 A circular conducting loop is stretched so its radius is $r(t) = r_0(1 + \alpha t)$, while a uniform field into the page is $B(t) = B_0(1 - \alpha t)$, where $\alpha > 0$ and $0 < t < 1/\alpha$. Neglect self-inductance. Viewed as in the diagram, what are the induced-current senses immediately after t=0 and at $t = 1/(2\alpha)$, respectively?

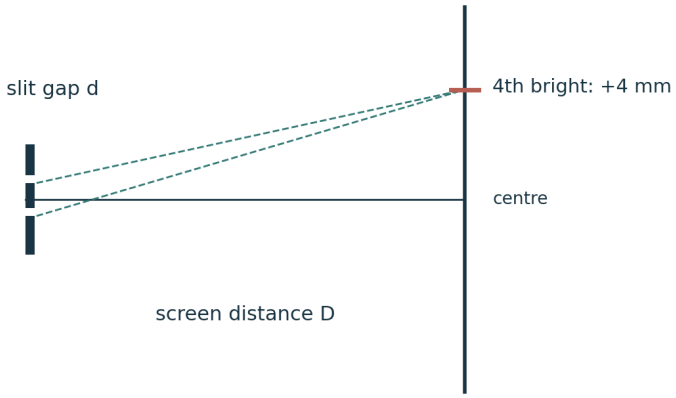
Field and loop area both change



A) Clockwise; clockwise
B) Clockwise; counterclockwise
C) Counterclockwise; counterclockwise
D) Counterclockwise; clockwise

Q57 A light ray in glass of index 1.50 strikes a flat glass-water interface at 50° to the normal; water index is 1.33, and a parallel water-air interface follows. Which statement is correct? Use air index 1.00.
A) Total internal reflection occurs immediately at glass-water
B) The ray enters water, then undergoes total internal reflection at water-air
C) The ray exits into air without reflection
D) The ray is normal to the second boundary

Q58 The figure shows a double-slit screen with fourth bright fringe 4 mm above the centre when slit separation is d and screen distance is D. If separation becomes 2d and screen distance becomes 1.5D at the same wavelength, where is that fourth fringe now?



A) 3 mm above
B) 2 mm above
C) 4 mm above
D) 6 mm above

Q59 A hydrogen atom is known to finish an absorption event in level n=3. The absorbed photon has energy about 1.89 eV. Which initial bound level was occupied? Use $E_n = -13.6/n^2$ eV.
A) 1
B) 2
C) 3
D) 4

Q60 A radioactive detector measures 50 counts/min after 12 h. Background is a constant 10 counts/min, and the source half-life is 4 h. What total count rate did it measure at the start?
A) 170 counts/min
B) 320 counts/min
C) 400 counts/min
D) 330 counts/min

Answer key and worked solutions

Every solution explains the selected choice and the alternative errors.

Biology

Q1 / A algal pigment and reserve product

Answer: (A) Rhodophyceae.

Why: Red algae combine those accessory pigments with floridean starch. Pigment and reserve location are independent evidence; aquatic habitat or visible colour alone cannot establish the class. The combination is diagnostic because green and brown algae use different reserve materials even when their habitats overlap.

Why not the others: (B) has fucoxanthin and laminarin/mannitol; (C) has a/b pigments and chloroplast starch; (D) is not this algal class.

Key point: Combine pigment and storage evidence.

Q2 / B leaf mesophyll distribution

Answer: (B) Dorsiventral dicot leaf.

Why: Palisade mesophyll near the illuminated upper surface and spongy mesophyll with air spaces beneath create a differentiated upper/lower pattern. Orientation in the diagram is essential to infer dorsiventral anatomy.

Why not the others: (A) lacks this strong two-layer differentiation; (C) lacks photosynthetic leaf mesophyll; (D) concerns a stem centre.

Key point: Use tissue identity and position together.

Q3 / C mitotic checkpoint

Answer: (C) Metaphase to anaphase.

Why: The replicated DNA signal shows S phase finished. An unsatisfied kinetochore attachment checkpoint delays anaphase, keeping sister chromatids together despite their doubled DNA content. Together the two observations locate the block.

Why not the others: (A) precedes replication; (B) concerns S completion already observed; (D) follows segregation.

Key point: Pair DNA content with chromatid behaviour.

Q4 / D light-response limiting factor

Answer: (D) The limiting factor changed.

Why: At dim light, light-driven ATP and reducing power constrain fixation. More light eases that constraint, so additional CO_2 now raises carbon assimilation. The controlled response identifies a condition-dependent limiting factor.

Why not the others: (A) contradicts fixation; (B) is unsupported; (C) cannot explain the light/CO_2 interaction.

Key point: Infer limiting factors from controlled response changes.

Q5 / A immune-related physiology

Answer: (A) IgE-mediated mast-cell degranulation.

Why: Prior exposure leaves specific IgE on mast cells. Re-exposure cross-links it and rapidly releases mediators that constrict airways and increase mucus. The minutes-long onset rules out building an entirely new adaptive response.

Why not the others: (B) a naive primary B-cell response cannot mature within minutes; (C) erythrocytes transport oxygen rather than making antibodies; (D) platelets function in haemostasis rather than secreting immunoglobulins.

Key point: Use timing and sensitisation to identify immediate allergy.

Q6 / B embryo-sac development

Answer: (B) Eight.

Given: One surviving megaspore divides three times.

Formula/Principle: Each mitosis doubles nuclei.

Step 1: 1→2→4→8.

Step 2: Degenerated megaspores do not contribute.

Why: Meiosis produces the initial four, but the common monosporic sac develops from one functional megaspore. Counting all four products would merge separate developmental lineages. The functional megaspore remains haploid during the mitoses; nuclear number changes without a new meiotic event or contribution from the discarded products.

Why not the others: (A) stops after only two mitoses; (C) assumes a fourth division; (D) assumes a fifth division of the functional megaspore.

Key point: Follow only the functional megaspore.

Q7 / C mutation and disorder

Answer: (C) 2/3.

Given: Aa×Aa yields AA:Aa:aa =1:2:1.

Formula/Principle: Condition on the three unaffected genotype outcomes.

Step 1: Aa probability 1/2; unaffected probability 3/4.

Step 2: Ratio (1/2)/(3/4)=2/3.

Why: Being healthy excludes aa but does not distinguish AA and Aa. Two of the three unaffected weighted outcomes carry one recessive allele. The condition is applied to one sibling's genotype distribution, independently of whether another child in the same family is affected.

Why not the others: (A) is unconditional AA probability; (B) unconditional carrier probability; (D) unaffected probability.

Key point: Recalculate within the observed phenotype.

Q8 / D Parasite life-cycle intervention and transmission

Answer: (D) The two interventions separate blood-stage disease from onward transmission.

Why: A feeding mosquito acquires sexual stages from human blood, whereas repeated asexual development in red blood cells is associated with the blood-stage attacks. Removing gametocytes therefore affects the mosquito-acquisition route without immediately removing all asexual parasites; removing asexual stages has the complementary immediate effect. The word initially matters because later population dynamics can change both outcomes.

Why not the others: (A) wrongly grants P an action against asexual blood stages; (B) wrongly grants Q an immediate action against gametocytes; (C) contradicts the specified human blood-stage targets.

Key point: Locate an intervention within the parasite life cycle before predicting disease and trans-

mission effects.

Q9 / A vector selection and marker

Answer: (A) Marker retention and possible insertional disruption.

Why: Antibiotic survival selects a functional resistance marker. Colour loss suggests interruption of the independent reporter, but sequencing/digest is needed to confirm insert identity and direction. Selection and screening answer different questions.

Why not the others: (B) contradicts survival under antibiotic selection; (C) contradicts the functional resistance marker needed for survival; (D) claims insert orientation from a colour readout that only screens reporter disruption.

Key point: Separate selectable marker from insertion screen.

Q10 / B competition and resource

Answer: (B) Competition.

Why: B's addition harms A under a shared limiting resource while environmental controls stay fixed. Resource competition is supported; the setup does not show B physically eating A. A controlled introduction supports an interaction effect, while the shared limiting seeds supply a plausible mechanism for the observed decline.

Why not the others: (A) predicts benefit; (C) predicts no effect on A; (D) needs direct consumption evidence.

Key point: Interpret a controlled population response with resource overlap.

Q11 / C Hardy-Weinberg assumption

Answer: (C) 0.42.

Given: $q^2=0.09$.

Formula/Principle: $q=0.3$, $p=0.7$ and heterozygotes $2pq$.

Step 1: Take the square root to recover q.

Step 2: $2(0.7)(0.3)=0.42$.

Why: The recessive phenotype is not the carrier frequency. Random mating and equilibrium assumptions permit genotype frequencies from allele frequencies. The equilibrium model would need reconsideration if selection, migration or nonrandom mating disturbed allele and genotype relationships.

Why not the others: (A) repeats aa; (B) misses factor two; (D) is p^2.

Key point: Use 2pq after deriving allele frequencies.

Q12 / D evolutionary selection evidence

Answer: (D) Differential inherited survival.

Why: Pre-existing variation plus colour-dependent predation and heritability changes allele frequency through reproduction. No directed environmental mutation is needed; the observation spans generations rather than individual transformation. The environmental change changes relative fitness; it does not imply that individual insects acquire an adaptive allele in response.

Why not the others: (A) is plasticity; (B) removes fitness differences; (C) removes heritable response.

Key point: Selection sorts inherited variants through differential fitness.

Q13 / A Cardiac output coupled to oxygen extraction

Answer: (A) Cardiac-output ratio 2; oxygen-consumption ratio 4.

Given: The oxygen-content difference is measured per litre, so cardiac output must first be expressed in L min^{-1}.

Formula/Principle:

$$Q = \text{HR} \times \text{SV}, \quad \dot{V}_{\text{O}_2} = Q(C_a - C_v).$$

Step 1:

$$Q_r = 75(0.080) = 6, \quad Q_e = 120(0.100) = 12 \ \text{L}\,\text{min}^{-1}.$$

Step 2:

$$\dot{V}_r = 6(200 - 150) = 300, \quad \dot{V}_e = 12(200 - 100) = 1200 \ \text{mL}\,\text{O}_2\,\text{min}^{-1}.$$

Step 3:

$$Q_e/Q_r = 2, \qquad \dot{V}_e/\dot{V}_r = 4.$$

Why: Oxygen delivery to tissues depends both on blood flow and on how much oxygen each litre loses across the tissues. Doubling flow while doubling extraction therefore quadruples the measured consumption in this steady-state model.

Why not the others: (B) ignores the extraction change; (C) and (D) use neither the measured output ratio nor the corresponding oxygen balance.

Key point: Combine circulation with oxygen extraction, maintaining volume units.

Q14 / B biomolecule linkage

Answer: (B) Primary structure.

Why: Peptide bonds make the covalent amino-acid chain. Cleaving them fragments its sequence, although higher structures may also collapse. The direct target is the backbone linkage, not a secondary hydrogen bond.

Why not the others: (A) describes assembly of several chains, whereas the enzyme cuts within a chain; (C) describes helix-stabilising hydrogen bonds rather than peptide bonds; (D) is a nucleic-acid linkage.

Key point: Identify the defining linkage of each structural level.

Q15 / C Calvin-cycle carbon tracing

Answer: (C) Ten.

Given: Twelve G3P of three carbons each, two exported.

Formula/Principle: Carbon conservation for regeneration.

Step 1: Return $12-2=10$ molecules.

Step 2: Ten·3=30 carbons equal six·5 in RuBP.

Why: Export uses two three-carbon units; the remainder recycles rather than forming one RuBP per G3P. The carbon check validates the stoichiometric partition. The thirty retained carbon atoms are reorganised into six five-carbon acceptors rather than being fixed anew in this regeneration phase.

Why not the others: (A) returns only 18 carbon atoms; (B) returns 24, also short of the 30 needed; (D) returns all 36 and leaves none for the exported sugar.

Key point: Track carbon through export and regeneration.

Chemistry

Q16 / B resonance contributor validity

Answer: (B) Nuclear positions and connectivity.

Why: Resonance moves electrons within the same atomic framework. Double-bond placement and individual formal charges vary; the actual hybrid has equivalent C–O bonds, not three exchanging molecules. Each contributor preserves the same total charge and electron count; only the distribution of bonding and nonbonding electrons differs.

Why not the others: (A) double-bond placement changes between contributors; (C) an individual oxygen's formal charge also changes; (D) incorrectly says total electron count changes, although it remains fixed.

Key point: Resonance moves electrons, not nuclei.

Q17 / A geometrical isomer count

Answer: (A) Cis-trans geometrical isomers.

Why: Ligands and connectivity match, but adjacency versus opposition is unchanged by rotating a square. X is cis and Y trans; no atom or ligand changes identity. Both square-planar drawings have four coordination sites, so the positional distinction is real even though the formula is identical.

Why not the others: (C) invokes mirror chirality; (B) changes coordination count; (D) cannot map an edge to a diagonal under rotation.

Key point: Compare relative positions in a fixed coordination geometry.

Q18 / D variable oxidation-state stability

Answer: (D) Medium changes redox product.

Why: Mn(VII) accepts different numbers of electrons and protons under the two conditions, ending as Mn(II) or Mn(IV). Balanced half-reactions explicitly depend on proton availability. Changing acidity changes proton availability in the reduction half-reaction, which changes how far manganese is reduced.

Why not the others: (A) denies variable states; (B) violates elemental identity; (C) contradicts MnO_2 production.

Key point: Specify medium for transition-metal redox.

Q19 / A effective nuclear charge trend

Answer: (A) Greater effective attraction within one shell.

Why: Added protons increase nuclear pull while valence electrons stay in n=2 and shielding does not fully compensate. Their average distance decreases. The broad trend need not be exactly smooth for all radius definitions.

Why not the others: (B) describes going down groups; (C) contradicts atomic numbers; (D) predicts no contraction.

Key point: Compare charge and shell number together.

Q20 / B quantum-number exclusions

Answer: (B) Row C.

Why: At n=3, l can be 0,1,2 only; l=3 violates $l \le n-1$. Each other row has allowed l, m_l in $[-l,l]$, and spin $\pm 1/2$. This first restriction is enough to reject C.

Why not the others: (A) is valid 3d; (C) valid 2p; (D) valid 4s.

Key point: Test quantum-number bounds in sequence.

Q21 / C gas stoichiometry under stated conditions

Answer: (C) 2.0 L.

Given: Reactant volume ratio 2:1 matches stoichiometry.

Formula/Principle: $2\,H_2 + O_2 \longrightarrow 2\,H_2O\,(g)$ and equal-condition volumes follow moles.

Step 1: Neither reactant remains.

Step 2: Two H_2 volumes form two vapour volumes.

Worked derivation:

$$2\,H_2 + O_2 \longrightarrow 2\,H_2O\,(g), \quad \frac{V_{H_2O}}{V_{H_2}} = \frac{2}{2} = 1 \Rightarrow V_{H_2O} = 2.0\ \text{L}.$$

Why: Avogadro's law applies only because product stays gaseous at the stated conditions. Condensed water would not have this volume ratio. The 2:1 reactant ratio matches the balanced equation, and the gaseous product coefficient is two, so no limiting-reagent correction is needed.

Why not the others: (A) halves product coefficient; (B) averages reactants; (D) adds their volumes.

Key point: Use gas ratios for gaseous products.

Q22 / D electrophile nucleophile selectivity

Answer: (D) Proton.

Why: The electron-rich alkene π bond attacks H^+, yielding the more stable carbocation; bromide attacks later. With no peroxide, a radical-chain pathway is not the proposed mechanism. The first step creates an electron-deficient carbon intermediate, which is why the bromide ion's role belongs to the subsequent step.

Why not the others: (A) is a nucleophile; (B) donates the electron pair; (C) belongs to another mechanism.

Key point: Locate the first electron-pair acceptor.

Q23 / A addition regioselectivity

Answer: (A) 2-bromopropane then 1-bromopropane.

Why: Ionic protonation favours a secondary carbocation and Markovnikov product in X. Radical Br addition in Y favours the secondary carbon radical, placing Br on the terminal carbon. Same reactants but changed initiator distinguish mechanisms.

Why not the others: (B) reverses pathways; (C) ignores peroxide; (D) assigns radical regiochemistry to both.

Key point: Predict regioselectivity from intermediate stability.

Q24 / B SN1 versus SN2 outcome

Answer: (B) Tertiary SN1; methyl SN2.

Why: Tertiary carbocation formation is comparatively stabilised and backside access crowded. Methyl cannot form a stable free carbocation but readily allows backside attack. Substrate geometry and cation stability both matter. A polar protic medium stabilises ions but cannot overcome the absence of a stable methyl carbocation; steric crowding favours the opposite route at tertiary carbon.

Why not the others: (A) requires methyl cation; (C) reverses trends; (D) wrongly excludes methyl substitution.

Key point: Compare steric access with cation stability.

Q25 / C alcohol oxidation selectivity

Answer: (C) Aldehyde versus ketone.

Why: Primary alcohol oxidation retains a hydrogen at the new carbonyl carbon, producing an aldehyde; secondary gives a ketone. Typical aldehydes reduce Tollens silver, ordinary ketones do not. Stopping before further aldehyde oxidation is essential.

Why not the others: (A) misclassifies primary; (B) misclassifies secondary; (D) names unrelated products.

Key point: Trace the original alcohol-bearing carbon.

Q26 / D reducing sugar before-after hydrolysis

Answer: (D) Free anomeric centres after hydrolysis.

Why: The intact glycosidic link joins both anomeric carbons, preventing an open-chain reducing form. Cleavage produces glucose and fructose; under alkaline testing, fructose can tautomerise and reduce reagent too. The before/after test isolates what the glycosidic bond masks; it does not mean sucrose lacked carbohydrate functional groups.

Why not the others: (A) reverses cleavage effect; (B) confuses proteins; (C) contradicts composition.

Key point: Check whether an anomeric carbon is free.

Q27 / C weak-acid dissociation

Answer: (C) Approximately 3.

Given: Ka≪C.

Formula/Principle: $[H^+] \simeq \sqrt{K_a C}$.

Step 1: $\sqrt{(10^{-5}\cdot 10^{-1})} = 10^{-3}$ M.

Step 2: pH=3; dissociation fraction $10^{-3}/0.10 = 1\%$, validating the approximation.

Worked derivation:

$$[\mathrm{H}^+] \simeq \sqrt{10^{-5}(10^{-1})} = 10^{-3}\ \mathrm{M}, \quad pH = 3, \quad \alpha = 10^{-3}/0.10 = 0.01.$$

Why: Mass action gives $x^2/(C-x)=Ka$; replacing $C-x$ by C must be checked afterward. A strong-acid calculation would assume full ionisation. A one-percent dissociation is small relative to the initial concentration, so the final numerical estimate is self-consistent.

Why not the others: (A) assumes full ionisation; (B) miscalculates root; (D) equates pH with pKa.

Key point: Verify approximations quantitatively.

Q28 / D van Hoff factor inference

Answer: (D) Two.

Given: Same Kf and molality; glucose i=1.

Formula/Principle: $\Delta T_f = iK_f m$.

Step 1: Ratio of depressions equals i/1=2.

Step 2: Complete ideal dissociation into two particles gives i=2.

Worked derivation:

$$\frac{\Delta T_{f,\text{solute}}}{\Delta T_{f,\text{glucose}}} = \frac{iK_f m}{1K_f m} = 2 \Rightarrow i = 2.$$

Why: Only effective particle number remains after solvent and molality cancel. Incomplete dissociation could complicate a real sample, but the ideal stipulation fixes the interpretation. The observation fixes effective particle number under the specified ideal complete-dissociation model, independent of the solute's identity.

Why not the others: (A) predicts the same depression as glucose; (B) predicts three times glucose;

(C) predicts four times glucose.

Key point: Infer particle count from a colligative ratio.

Q29 / A cell polarity from quotient

Answer: (A) 1.0408 V.

Given: n=2, log Q=2.

Formula/Principle: Insert into supplied Nernst relation.

Step 1: Correction (0.0592/2)(2)=0.0592 V.

Step 2: Subtract from 1.10 V.

Worked derivation:

$$E = 1.10 - \frac{0.0592}{2}\log_{10}(100) = 1.10 - 0.0592 = 1.0408 \text{ V}.$$

Why: Product-rich Q>1 lowers forward cell voltage; adding the correction reverses the physical direction. Electron count divides the correction. The correction has voltage units and is positive before subtraction because log100=2, so the result must remain just below E°.

Why not the others: (B) adds; (C) doubles correction; (D) ignores Q.

Key point: Check Nernst sign against Q.

Q30 / B enthalpy versus entropy sign

Answer: (B) Sufficiently high temperature.

Worked derivation:

$$\Delta G = \Delta H - T\Delta S < 0 \iff T > \Delta H/\Delta S \quad (\Delta S > 0).$$

Why: The unfavourable enthalpy is offset increasingly by the favourable entropy term as T rises. Positive ΔS alone does not determine low-temperature spontaneity. At the crossover temperature ΔG=0; below it the positive enthalpy dominates, and above it the entropy term dominates.

Why not the others: (A) reverses trend; (C) ignores ΔH; (D) ignores entropy.

Key point: Compare both signed Gibbs terms.

Mathematics

Q31 / A ellipse focus-directrix

Answer: (A) e=3/5 and x=25/3.

Given: a=5, c=3.

Formula/Principle: e=c/a; right directrix x=a/e.

Step 1: e=3/5.

Step 2: x=5/(3/5)=25/3.

Worked derivation:

$$a = 5,\ c = 3, \quad e = \frac{c}{a} = \frac{3}{5}, \quad x_{directrix} = \frac{a}{e} = \frac{25}{3}.$$

Why: An ellipse has e<1; its directrix must lie beyond the vertex. Confusing the focal location with directrix fails this geometric check. The right directrix is located at a/e, not at the focus; because e is less than one, a/e exceeds the semimajor axis.

Why not the others: (B) gives e>1; (C) inverts formula; (D) uses semiminor length as c.

Key point: Distinguish focal distance from directrix distance.

Q32 / D parabola normal

Answer: (D) (18,−12).

Given: The first intersection is (2,4).

Formula/Principle: Implicit differentiation finds the normal; substitute its line into the parabola.

Step 1: 2yy′=8 gives tangent slope 1 at (2,4), so normal slope −1 and line y=6−x.

Step 2: Substitute $(6-x)^2=8x$, or $x^2-20x+36=0=(x-2)(x-18)$.

Step 3: The other x is 18 and y=6−18=−12.

Worked derivation:

$$y' = 4/y = 1, \quad y = 6 - x, \quad (6-x)^2 = 8x \Rightarrow (x-2)(x-18) = 0 \Rightarrow (18, -12).$$

Why: The derivative determines the normal at the specified point; its second crossing is obtained only after solving the line-parabola system. The known root x=2 is discarded as the first intersection, and the second point satisfies both equations.

Why not the others: (A) has wrong sign of y and fails the normal; (B) is the normal's x-intercept but not on the parabola; (C) reuses x=2 with the wrong y for that point.

Key point: Solve the normal and original curve simultaneously.

Q33 / A partial fractions with limits

Answer: (A) ln(4/3).

Given: Distinct linear factors.

Formula/Principle: $1/[(x+1)(x+2)] = 1/(x+1) - 1/(x+2)$.

Step 1: Antiderivative ln[(x+1)/(x+2)].

Step 2: Upper ln(2/3) minus lower ln(1/2) gives ln(4/3).

Worked derivation:

$$\left[\ln \frac{x+1}{x+2}\right]_0^1 = \ln \frac{2}{3} - \ln \frac{1}{2} = \ln \frac{4}{3}.$$

Why: Both limits contribute. The positive integral must produce a positive logarithm, providing a quick sign check. The logarithm of a ratio combines endpoint terms and keeps the expression dimensionless, consistent with the positive area under the integrand.

Why not the others: (B) loses a factor; (C) omits one fraction; (D) treats integrand as constant.

Key point: Evaluate both endpoints after partial fractions.

Q34 / C Absolute-value integral and sign splitting

Answer: (C) 1.

Given: $|x| = -x$ for negative x, and equals x for nonnegative x.

Formula/Principle: Split the integral where the absolute-value formula changes.

Step 1:

$$\int_{-1}^{0} (-x)\, dx = \frac{1}{2}.$$

Step 2:

$$\int_0^1 x\, dx = \frac{1}{2}, \qquad I = \frac{1}{2} + \frac{1}{2} = 1.$$

Worked derivation:

$$I = 2\int_0^1 x\, dx = 2\left[\frac{x^2}{2}\right]_0^1 = 1.$$

Why: Absolute value makes both triangular regions positive. The integrand is even, so cancellation of an odd function cannot be applied.

Why not the others: (A) wrongly replaces absolute value by x; (B) counts only one half; (D) uses a rectangle instead of two triangles.

Key point: Resolve absolute values before using symmetry.

Q35 / C implicit tangent

Answer: (C) 7/2.

Given: (1,2) satisfies the curve.

Formula/Principle: Product-rule implicit differentiation.

Step 1: $2x + y + (x + 2y)y' = 0$ gives slope $-4/5$.

Step 2: In y−2=−(4/5)(x−1), set y=0, giving x=7/2.

Worked derivation:

$$y' = -\frac{2x+y}{x+2y}\Big|_{(1,2)} = -\frac{4}{5}, \quad -2 = -\frac{4}{5}(x-1) \Rightarrow x = \frac{7}{2}.$$

Why: Differentiating xy needs both x y′ and y. The intercept belongs to the tangent line, not another intersection of the original curve. At the specified point the denominator x+2y is nonzero, ensuring that the local tangent has a finite slope.

Why not the others: (A) drops the point's x offset; (B) rounds; (D) uses the wrong slope.

Key point: Keep the product-rule term.

Q36 / D injection versus surjection

Answer: (D) Surjective but not injective.

Worked derivation:

$$f(0) = f(\sqrt{3}) = f(-\sqrt{3}) = 0, \quad \lim_{x \to \pm\infty} f(x) = \pm\infty.$$

Why: Three inputs have output zero, disproving one-to-one behaviour. Continuity and opposite unbounded end behaviour ensure every real value is reached by the intermediate-value theorem. The repeated-output counterexample is sufficient to disprove injection, whereas surjection needs the separate global continuity and endpoint argument.

Why not the others: (A) falsely claims injection despite three inputs giving zero; (B) repeats that injection error and denies the full range; (C) correctly denies injection but overlooks surjectivity.

Key point: Test injection and surjection separately.

Q37 / C adjoint identity

Answer: (C) 2.

Given: Matrix order three and det A=−2.

Formula/Principle: det(A⁻¹)=1/det A; det(adj B)=(det B)^(n−1); det(cB)=c^n det B.

Step 1: det(A⁻¹)=−1/2.

Step 2: det(adj(A⁻¹))=(−1/2)²=1/4.

Step 3: Multiplication of a 3×3 matrix by 2 multiplies determinant by 2³=8, giving 8/4=2.

Worked derivation:

$$\det(2\,\mathrm{adj}(A^{-1})) = 2^3(\det A^{-1})^{3-1} = 8(-\tfrac{1}{2})^2 = 2.$$

Why: The inverse changes the determinant reciprocally, the adjugate contributes an order-dependent square, and the outer scalar acts on all three rows. Losing any of these stages changes sign or magnitude.

Why not the others: (A) copies the original signed determinant; (B) stops at the inverse magnitude; (D) applies only the outer scalar factor.

Key point: Apply inverse, adjugate and scalar identities in order.

Q38 / D independent versus exclusive events

Answer: (D) 3/5 and not independent.

Given: Disjoint events of positive probability.

Formula/Principle: Union is 5/6; independence requires intersection 1/6.

Step 1: Conditional probability (1/2)/(5/6)=3/5.

Step 2: Actual intersection 0 differs from (1/2)(1/3)=1/6.

Worked derivation:

$$P(A \mid A \cup B) = \frac{1/2}{5/6} = \frac{3}{5}, \quad P(A \cap B) = 0 \neq \frac{1}{6}.$$

Why: The condition changes the denominator. Positive-probability disjoint events cannot also be independent because independence predicts overlap. Mutual exclusion answers one joint-probability question, whereas independence imposes a different equality; confusing them loses the conditioning structure.

Why not the others: (A) ignores condition; (B) misclassifies independence; (C) uses wrong denominator.

Key point: Calculate both conditional and intersection tests.

Q39 / A inverse-value branch

Answer: (A) −π/3.

Given: Tangent repeats every π.

Formula/Principle: Shift input into principal inverse range.

Step 1: 2π/3−π=−π/3.

Step 2: tan(2π/3)=−√3, whose principal arctan is −π/3.

Worked derivation:

$$\arctan(\tan(2\pi/3)) = \arctan(-\sqrt{3}) = -\pi/3.$$

Why: The inverse returns its specified branch, not necessarily the original angle. Periodicity supplies equivalent angles but the range chooses one. The selected output is the unique angle in the principal interval with tangent −√3, even though adding integer multiples of π gives the same tangent.

Why not the others: (B) is the original angle outside the principal range; (C) has positive tangent rather than negative; (D) also lies outside the principal range.

Key point: Check inverse-trigonometric principal ranges.

Q40 / B skew-line distance

Answer: (B) 1/√2.

Given: Directions (1,0,0),(0,1,1); point difference (0,1,2).

Formula/Principle: Project point difference on their common normal.

Step 1: Cross product (0,−1,1), norm √2.

Step 2: Dot product (0,1,2)·(0,−1,1)=1.

Worked derivation:

$$D = \frac{|(0,1,2)\cdot(0,-1,1)|}{\sqrt{2}} = \frac{1}{\sqrt{2}}.$$

Why: Distance between arbitrary listed points is not the minimum; only the perpendicular connector is. The cross product is perpendicular to both directions; dividing by its magnitude converts the scalar triple product into an actual distance.

Why not the others: (A) omits normalisation; (C) multiplies by norm; (D) uses one coordinate alone.

Key point: Project onto the cross-product direction.

Q41 / D dot-product projection

Answer: (D) (2/9)b.

Given: a·b=2 and |b|²=9.

Formula/Principle: $\text{proj}_b\, a = (a \cdot b/|b|^2)b$.

Step 1: Coefficient 2/9.

Step 2: Vector components (4/9,−2/9,4/9).

Worked derivation:

$$\text{proj}_b\, a = \frac{2}{9}(2,-1,2) = (4/9,-2/9,4/9).$$

Why: Scalar projection is 2/3; vector projection divides by |b| squared and points in b's direction. The projected vector has the same sign pattern as b because the dot product is positive, which checks the result's direction.

Why not the others: (A) confuses scalar coefficient; (B) points along a; (C) inverts scale.

Key point: Distinguish scalar from vector projection.

Q42 / B conjugate polynomial condition

Answer: (B) −4.

Given: Real coefficients require root 1−i too.

Formula/Principle: Multiply linear root factors.

Step 1: Conjugate pair gives (x−1)²+1=x²−2x+2.

Step 2: Multiply by x−2; constant (−2)·2=−4.

Worked derivation:

$$p(x) = (x-2)((x-1)^2+1) = x^3 - 4x^2 + 6x - 4.$$

Why: Omitting the conjugate leaves only a quadratic, incompatible with a real monic cubic. The real-coefficient condition is essential; without it, a nonreal root alone would not force its conjugate.

Why not the others: (A) omits pair contribution; (C) loses sign; (D) is another coefficient.

Key point: Complete complex conjugate pairs.

Q43 / A strict interval root test

Answer: (A) −1<k<0.

Given: Roots 1±√(1+k).

Formula/Principle: Distinct real requires 1+k>0; lower root positive requires √(1+k)<1.

Step 1: k>−1.

Step 2: k<0; upper root then <2<3.

Worked derivation:

$$r_\pm = 1 \pm \sqrt{1+k}; \quad k > -1, \quad 1 - \sqrt{1+k} > 0 \Rightarrow k < 0.$$

Why: k=−1 repeats a root, k=0 puts one on excluded endpoint. The upper bound follows automatically once lower bound holds. Testing the two boundary values explicitly exposes why closed inequalities would incorrectly include an endpoint root or a repeated root.

Why not the others: (B) includes a repeated-root case and a root on x=0; (C) makes the lower root negative; (D) likewise places the lower root outside (0,3).

Key point: Check distinctness and strict bounds.

Q44 / B middle-term ratio

Answer: (B) 18/5.

Given: Indices 4,5 for n=9 and 4 for n=8.

Formula/Principle: Binomial coefficients C(n,k).

Step 1: C(9,4)=C(9,5)=126; sum252.

Step 2: C(8,4)=70; ratio252/70=18/5.

Worked derivation:

$$\frac{\binom{9}{4}+\binom{9}{5}}{\binom{8}{4}} = \frac{126+126}{70} = \frac{18}{5}.$$

Why: Odd powers have two central terms; using one halves the numerator. The equality of the two odd-power central coefficients follows from binomial symmetry, but both still contribute to the requested sum.

Why not the others: (A) uses one middle; (C) miscounts denominator; (D) rounds an exact ratio.

Key point: Determine middle indices first.

Q45 / C arithmetic-geometric progression

Answer: (C) 1793.

Given: Finite weighted geometric series.

Formula/Principle: Shift and subtract S from 2S.

Step 1: General identity $S_n = (n-1)2^n + 1$.

Step 2: n=8 gives 7·256+1=1793.

Worked derivation:

$$S_8 = \sum_{k=1}^{8} k2^{k-1} = (8-1)2^8 + 1 = 1793.$$

Why: The shifted subtraction cancels interior powers, leaving an endpoint term. Plain geometric summation omits k. The identity can be verified by shifting 2S and subtracting S, leaving one copy of each interior power and the final weighted endpoint.

Why not the others: (A) stops after seven weighted terms; (B) uses 6 rather than 7 as the endpoint coefficient in the finite-sum identity; (D) uses 8 rather than 7 for that coefficient.

Key point: Derive weighted sums by index shift.

Physics

Q46 / C vertical-motion turning time

Answer: (C) 3 s.

Given: Upward-positive initial velocity 20; first-stage change −10(1) m/s.

Formula/Principle: Apply $v = v_0 + at$ separately on each stage.

Step 1: At t=1 s, v=20−10=10 m s⁻¹ upward.

Step 2: At downward acceleration 5, an additional 10/5=2 s is needed to reach v=0.

Worked derivation:

$$v_1 = 20 - 10(1) = 10\,\mathrm{m,s^{-1}}, \quad \Delta t_2 = \frac{10}{5} = 2\,\mathrm{s}, \quad t_{turn} = 1 + 2 = 3\,\mathrm{s}.$$

Why: The changed acceleration begins after the first second, so using either value for the entire flight gives the wrong turning time. The velocity remains positive at the boundary, making the second stage necessary.

Why not the others: (A) assumes g=10 throughout; (B) averages accelerations without time weighting; (D) assumes g=5 throughout.

Key point: Carry boundary velocity into the next stage.

Q47 / A coupled tension with support acceleration

Answer: (A) 4 N.

Given: Smooth horizontal floor; upward support acceleration changes vertical normal forces.

Formula/Principle: Horizontal acceleration of the two-block system $a_x = F/(m_1 + m_2)$, and string pulls the 2 kg block.

Step 1: $a_x = 10/(2+3) = 2\,\mathrm{m,s^{-2}}$.

Step 2: $T = m_1 a_x = 2(2) = 4\,\mathrm{N}$.

Worked derivation:

$$a_x = \frac{10}{5} = 2\,\mathrm{m,s^{-2}}, \qquad T = (2\,\mathrm{kg})a_x = 4\,\mathrm{N}.$$

Why: Elevator acceleration acts vertically, so its pseudo-force alters normal reactions but not the frictionless horizontal force balance. This setup tests which acceleration component belongs in the tension equation, unlike a vertical Atwood system.

Why not the others: (B) overlooks mass; (C) uses 3 kg for the pulled block's tension relation; (D) equates string tension to the entire applied force.

Key point: Resolve support motion and string force by direction.

Q48 / D spring-gravity conversion

Answer: (D) 0.20 m.

Given: Gravity acts over the initial fall plus spring compression x.

Formula/Principle: At maximum compression speed is zero, $\frac{1}{2}kx^2 = mg(h+x)$.

Step 1: $100x^2 = 10(0.20 + x)$, or $100x^2 - 10x - 2 = 0$.

Step 2: Factor $(10x-2)(10x+1) = 0$; physical positive root x=0.20 m.

Worked derivation:

$$100x^2 = 2 + 10x \Rightarrow (10x-2)(10x+1) = 0 \Rightarrow x = 0.20\,\mathrm{m}.$$

Why: Gravitational potential keeps decreasing while the spring compresses, so using only the 0.20 m free-fall height underestimates stored elastic energy. The negative algebraic root is not a compression.

Why not the others: (A) stops before force balance; (B) omits part of gravitational work; (C) stores too much spring energy.

Key point: Include descent after contact in energy conservation.

Q49 / B rolling energy partition

Answer: (B) $2\sqrt{5}\,\mathrm{m,s^{-1}}$.

Given: Pure rolling, $\omega = v/R$, height h=1.5 m.

Formula/Principle: $Mgh = \frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2 = \frac{3}{4}Mv^2$.

Step 1: $v^2 = (4/3)gh = (4/3)(10)(1.5) = 20$.

Step 2: Positive speed $v = \sqrt{20} = 2\sqrt{5}$.

Worked derivation:

$$Mgh = \frac{1}{2}Mv^2 + \frac{1}{4}Mv^2 = \frac{3}{4}Mv^2, \quad v^2 = \frac{4}{3}(10)(1.5) = 20.$$

Why: Some gravitational energy appears as rotation, so the translational speed is below the sliding

value $\sqrt{2gh} = \sqrt{30}$. The rolling constraint connects angular and linear speeds without specifying mass or radius.

Why not the others: (A) misallocates rotational energy; (C) assumes no rotational kinetic energy; (D) doubles the energy result.

Key point: Include rotation in rolling energy.

Q50 / C satellite period under radius change

Answer: (C) 8.

Given: Central mass unchanged; circular gravitational orbit.

Formula/Principle: $T = 2\pi\sqrt{r^3/(GM)}$.

Step 1: $T'/T = \sqrt{(4r)^3/r^3}$.

Step 2: $\sqrt{64} = 8$.

Worked derivation:

$$\frac{T'}{T} = \left(\frac{4r}{r}\right)^{3/2} = 4^{3/2} = 8.$$

Why: Gravity weakens with radius and the orbit circumference also grows; the combined circular-orbit dynamics produces the r^(3/2) period law. Holding g constant as at a planet's surface would be an invalid assumption for a radius change this large.

Why not the others: (A) takes only a square root; (B) scales period like radius; (D) squares radius.

Key point: Use the same central mass in Kepler's period scaling.

Q51 / B buoyancy under partial immersion

Answer: (B) 3/8.

Given: Weight $W = 0.8\rho_w gV$; buoyancy for half volume $B = 0.5\rho_w gV$.

Formula/Principle: Apparent weight $W_{app} = W - B$.

Step 1: $W_{app} = (0.8 - 0.5)\rho_w gV = 0.3\rho_w gV$.

Step 2: Divide by true weight 0.8ρgV to obtain 0.3/0.8=3/8.

Worked derivation:

$$\frac{W_{app}}{W} = \frac{0.8\rho gV - 0.5\rho gV}{0.8\rho gV} = \frac{3}{8}.$$

Why: If released, the block would float with 80% submerged, but the stated 50% is held by an external support. Its apparent weight is positive at that immersion; the free-floating fraction is a different condition.

Why not the others: (A) subtracts fractions after division wrongly; (C) assumes equal density; (D) adds rather than subtracts buoyancy.

Key point: Use displaced volume for buoyancy and the actual weight for the denominator.

Q52 / C first-law sign convention

Answer: (C) +250 J.

Given: Work by the gas is positive in the convention $\Delta U = Q - W_{by}$.

Formula/Principle: Add state-energy changes across stages.

Step 1: Expansion $\Delta U_1 = 300 - 100 = 200\,\text{J}$.

Step 2: Adiabatic compression Q=0, W_by=−50 J, so $\Delta U_2 = +50\,\text{J}$.

Worked derivation:

$$\Delta U_{total} = (300 - 100) + (0 - (-50)) = 250\,\text{J}.$$

Why: Compression does work on the gas and raises its internal energy even without heat transfer. A single consistent sign convention across both stages prevents mistakenly subtracting the com-

pression work. Adding stagewise internal-energy changes is legitimate because internal energy is a state function, even though heat and work depend on path.

Why not the others: (A) subtracts the 50 J compression input; (B) omits stage two; (D) adds expansion work instead of subtracting it.

Key point: State the work sign convention before combining stages.

Q53 / D dipole torque and energy

Answer: (D) Same endpoint torque magnitude; energy rises by pE.

Worked derivation:

$$|\tau_{60}| = |\tau_{120}| = pE\sin 60^\circ, \quad U_{60} = -\frac{pE}{2},\ U_{120} = +\frac{pE}{2}, \quad \Delta U = pE.$$

Why: Sine has the same value at supplementary angles, but cosine reverses sign. Moving the dipole away from alignment raises its field potential energy; comparing torque alone cannot tell whether energy increases or decreases.

Why not the others: (A) overlooks cosine change; (B) gives wrong torque and energy direction; (C) gets torque but reverses energy.

Key point: Use sine for torque and negative cosine for dipole energy.

Q54 / A plate-separation change

Answer: (A) Fixed-voltage energy halves, fixed-charge energy doubles.

Worked derivation:

$$C' = C/2; \quad U'_V = \tfrac{1}{2}C'V^2 = U_V/2, \qquad U'_Q = Q^2/(2C') = 2U_Q.$$

Why: The battery can exchange charge to maintain V, whereas disconnection fixes Q. Changing the same geometry therefore changes energy in opposite directions under the two boundary conditions. Mechanical work and battery exchange account for the different energy transfers.

Why not the others: (B) applies fixed-V rule twice; (C) reverses both constraints; (D) applies fixed-Q rule twice.

Key point: Identify what is held constant before using the capacitor energy formula.

Q55 / C parallel-wire force

Answer: (C) Factor 2.

Given: Initial product $I(2I) = 2I^2$; final product $(2I)(4I) = 8I^2$, separation 2d.

Formula/Principle: $F/L = \mu_0 I_1 I_2/(2\pi d)$.

Step 1: Current product rises by factor 4.

Step 2: Distance denominator rises by factor 2; net factor 4/2=2.

Worked derivation:

$$\frac{(F/L)'}{F/L} = \frac{(2I)(4I)/(2d)}{I(2I)/d} = 2.$$

Why: Each wire's current contributes linearly to the magnetic interaction, while separation contributes inversely. Same-direction attraction remains attraction; the question asks magnitude scaling, not force direction. Same-direction currents remain attractive after scaling; only the magnitude ratio changes in the calculation.

Why not the others: (A) accounts only for distance; (B) doubles just one current; (D) omits distance change.

Key point: Scale both currents and separation in one ratio.

Q56 / D Competing field and area changes reverse induced current

Answer: (D) Counterclockwise initially; clockwise at $1/(2\alpha)$.

Given: Define inward magnetic flux as positive; both the field and loop area change.

Formula/Principle:

$$\Phi(t) = B_0 \pi r_0^2 (1 - \alpha t)(1 + \alpha t)^2.$$

Step 1: Differentiate the complete product:

$$\frac{d\Phi}{dt} = B_0 \pi r_0^2 \alpha (1 + \alpha t)(1 - 3\alpha t).$$

Step 2: Initially this derivative is positive: inward flux increases despite the decreasing field. The induced field is outward, so current is counterclockwise.

Step 3: At $t = 1/(2\alpha)$, $1 - 3\alpha t = -1/2$, so inward flux decreases. The induced field is inward, so current is clockwise.

Worked derivation:

$$\operatorname{sgn} \dot{\Phi}(0^+) = +, \qquad \operatorname{sgn} \dot{\Phi}(1/(2\alpha)) = -.$$

Why: Lenz's law opposes the change of flux, not the change of field magnitude alone. The expanding area can initially dominate the declining field, and the competition reverses later in the stated interval.

Why not the others: (A) ignores initial area growth; (B) reverses both Lenz-law responses; (C) ignores the later sign reversal.

Key point: Differentiate the full field-times-area flux before applying a current-direction rule.

Q57 / B critical angle through layers

Answer: (B) It enters water but cannot pass into air.

Given: Parallel boundaries retain the same refracted angle within water.

Formula/Principle: Snell law and critical condition $n_{high} \sin \theta_c = n_{low}$.

Step 1: Glass-water: $1.50 \sin 50^\circ \approx 1.149 < 1.33$, so transmission into water is possible.

Step 2: For water-air transmission this same tangential quantity 1.149 would have to equal $1.00 \sin \theta_{air} \leq 1$, impossible.

Worked derivation:

$$n_g \sin 50^\circ \approx 1.149 < n_w = 1.33, \quad n_w \sin \theta_w = 1.149 > n_{air} = 1.00 \Rightarrow \text{TIR at water-air}.$$

Why: The first interface's critical threshold differs from the second. A thin intermediate layer can admit the ray yet still prevent exit through a parallel lower-index boundary; Snell's invariant makes the conclusion clear without calculating both angles.

Why not the others: (A) overlooks allowable glass-water transmission; (C) requires sine above one in air; (D) contradicts oblique incidence.

Key point: Compare conserved tangential optical quantity with each next medium.

Q58 / A slit separation and fringe shift

Answer: (A) 3 mm above centre.

Given: Fringe position $y_m = m\lambda D/d$; m and wavelength unchanged.

Formula/Principle: New-to-old position ratio $(D'/D)/(d'/d) = 1.5/2 = 3/4$.

Step 1: Apply factor to old position 4 mm.

Step 2: New y_4=(3/4)(4)=3 mm, same side of centre.

Worked derivation:

$$\frac{y_4'}{y_4} = \frac{1.5D/(2d)}{D/d} = \frac{3}{4}, \qquad y_4' = \frac{3}{4}(4\,\text{mm}) = 3\,\text{mm}.$$

Why: Increasing D spreads fringes while increasing d compresses them; both changes must be combined as a ratio. The order m=4 remains fixed, so the result is a shifted position, not a different fringe number.

Why not the others: (B) uses doubled d alone; (C) assumes cancellation; (D) uses increased D alone.

Key point: Scale fringe position by D/d.

Q59 / B energy-level absorption

Answer: (B) n=2.

Given: Final energy $E_3 = -13.6/9 \approx -1.51$ eV; photon energy positive 1.89 eV.

Formula/Principle: $E_{final} - E_{initial} = E_\gamma$.

Step 1: $E_{initial} \approx -1.51 - 1.89 = -3.40\,\text{eV}$.

Step 2: $-13.6/n^2 = -3.40$ gives n^2=4, n=2.

Worked derivation:

$$E_i = E_3 - E_\gamma = -\frac{13.6}{9} - 1.89 \approx -3.40\,\text{eV} = -\frac{13.6}{2^2}.$$

Why: Absorption raises energy toward zero, so the initial bound state must have a more negative energy than n=3. The photon energy is the difference, not the magnitude of either level alone.

Why not the others: (A) requires a much larger jump; (C) gives zero photon energy; (D) starts above n=3 and would emit rather than absorb.

Key point: Reconstruct initial energy by subtracting absorbed photon energy.

Q60 / D radioactivity count rate

Answer: (D) 330 counts/min.

Given: Three half-lives elapsed; measured late count includes 10 background.

Formula/Principle: Source rate after t is initial source rate times $(1/2)^{t/T_{1/2}}$.

Step 1: Late net source rate is 50−10=40 counts/min.

Step 2: Initial net is 40×2^3=320; add the same background 10 to get total 330.

Worked derivation:

$$R_{source,0} = (50-10)2^{12/4} = 40 \cdot 8 = 320, \quad R_{total,0} = 320 + 10 = 330\,\text{counts/min}.$$

Why: Background does not decay with the sample and must be removed before reversing three half-lives, then restored to the total reading. Multiplying the full measured 50 by eight would incorrectly amplify the background.

Why not the others: (A) reverses only two half-lives; (B) omits restored background; (C) multiplies background along with source.

Key point: Subtract constant background before half-life reconstruction.
